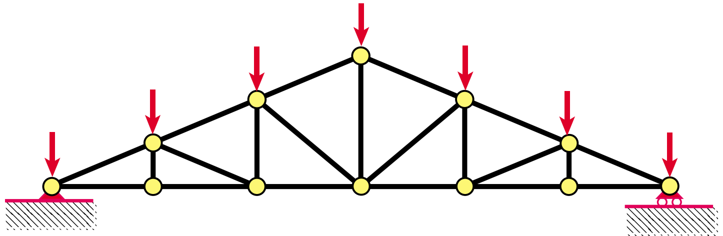
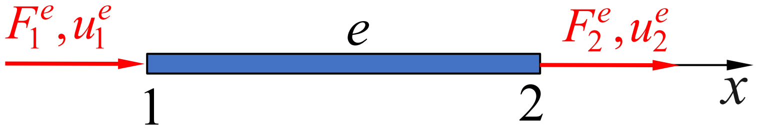
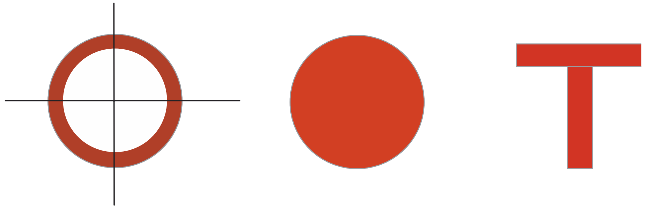
第2章
直接刚度法

2.2 单元刚度方程
Element Stiffness Equation





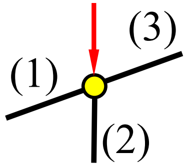


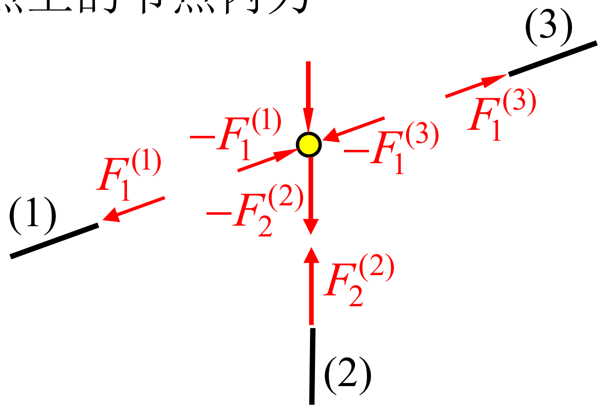
x: 从节点1指向节点2

u_I^e: 节点位移 ($I = 1,2$)

F_I^e: 作用在单元上的节点内力(指向x正方向时为正)

$-F_I^e$: 单元作用在节点上的节点内力



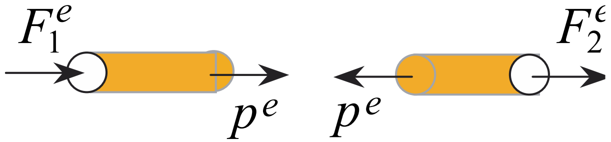


应变 $\varepsilon^e = \dfrac{\Delta^e}{l^e} = \dfrac{u_2^e - u_1^e}{l^e}$

$$= \underbrace{\frac{1}{l^e}\begin{bmatrix} -1 & 1 \end{bmatrix}}_{\boldsymbol{B}^e} \underbrace{\begin{bmatrix} u_1^e \\ u_2^e \end{bmatrix}}_{\boldsymbol{d}^e}$$

$$\boxed{\varepsilon^e = \boldsymbol{B}^e \boldsymbol{d}^e}$$

? 如何建立节点内力和节点位移之间的关系?



单元轴向内力:

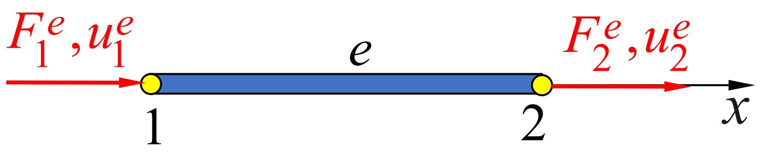
$$p^e = A^e \sigma^e \quad \text{(受拉为正)}$$
$$= A^e E^e \varepsilon^e \quad \text{(胡克定律)}$$

作用在单元上的节点内力:

$$F_1^e = -p^e = -A^e E^e \varepsilon^e = k^e (u_1^e - u_2^e)$$

$$F_2^e = p^e = k^e (u_2^e - u_1^e)$$

$$k^e = \frac{A^e E^e}{l^e}$$



$$F_1^e = k^e(u_1^e - u_2^e)$$

$$F_2^e = k^e(u_2^e - u_1^e)$$

$$\underbrace{\begin{bmatrix} F_1^e \\ F_2^e \end{bmatrix}}_{\boldsymbol{F}^e} = \underbrace{k^e \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}}_{\boldsymbol{K}^e} \underbrace{\begin{bmatrix} u_1^e \\ u_2^e \end{bmatrix}}_{\boldsymbol{d}^e}$$

$$\boldsymbol{K}^e = \frac{A^e E^e}{l^e}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$ — 单元刚度矩阵

$$\boldsymbol{F}^e = [F_1^e \ \ F_2^e]^{\mathrm{T}}$$ — 单元节点内力列阵

$$\boldsymbol{d}^e = [u_1^e \ \ u_2^e]^{\mathrm{T}}$$ — 单元节点位移列阵

$$\boxed{\boldsymbol{F}^e = \boldsymbol{K}^e \boldsymbol{d}^e}$$

单元刚度方程

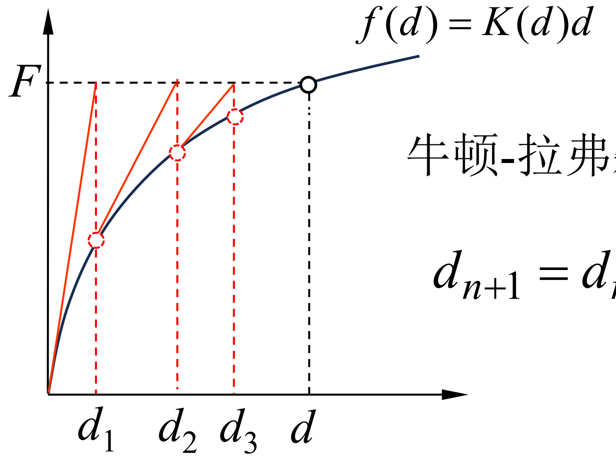
- ✧ 给出了单元节点内力列阵和节点位移列阵之间的关系
- ✧ 是一个线性方程
  - ○ 线弹性：胡克定理
  - ○ 小变形：轴向力和应力之间满足线性关系
  - ○ 小变形：应变和位移之间满足线性关系
- ✧ 对于 $u_1^e = u_2^e$ (刚体模态)，单元内力为零

本课程仅讨论线性问题！

非线性：几何非线性/材料非线性？

$$\boldsymbol{F}^e = \underbrace{\boxed{\boldsymbol{K}^e(\boldsymbol{d}^e)}}_{\text{(非常数)}}\boldsymbol{d}^e$$

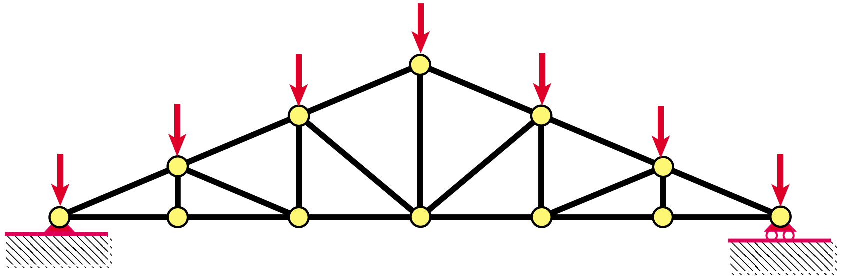
- 迭代求解
- 增量加载

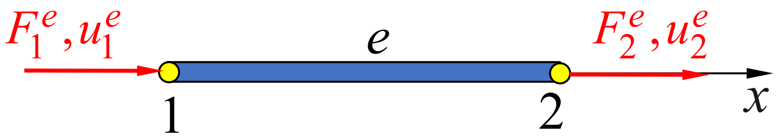


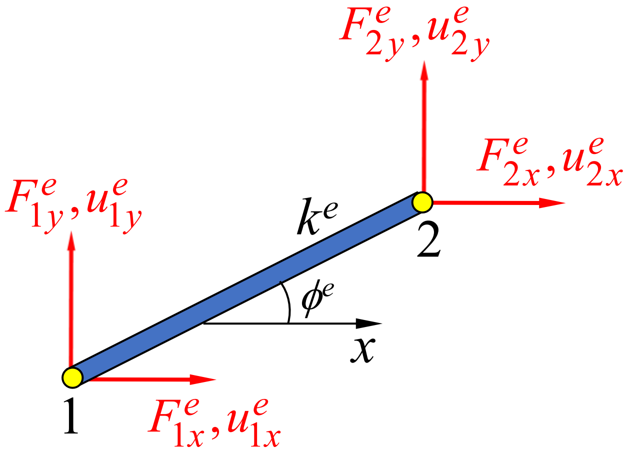
牛顿-拉弗森法

$$d_{n+1} = d_n + \frac{F - f(d_n)}{K(d_n)}$$

- 桁架是由若干个细长构件（称为杆单元）铰接而成的结构
- 杆单元在空间中可以沿任意方向
- 二维和三维空间中沿任意方向的杆单元的单元刚度矩阵？







$$\boldsymbol{F}^e = \boldsymbol{K}^e \boldsymbol{d}^e$$

$$\boldsymbol{d}^e = [u_1^e \;\; u_2^e]^{\mathrm{T}}$$

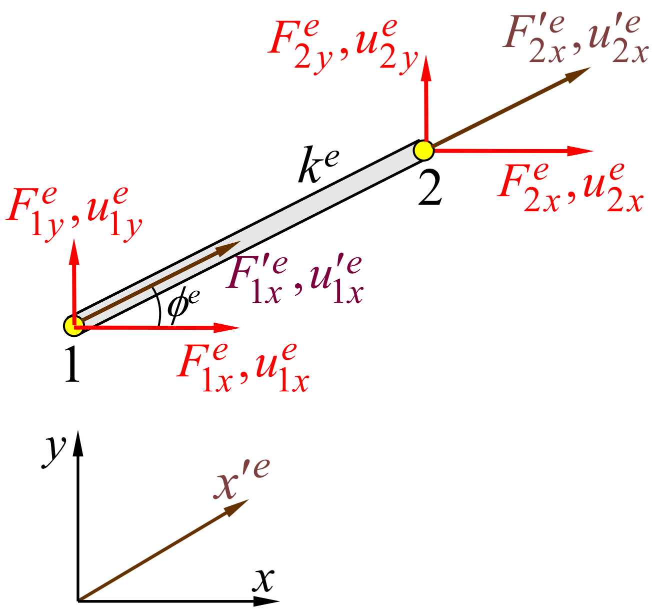
$$\boldsymbol{F}^e = [F_1^e \;\; F_2^e]^{\mathrm{T}}$$

?

$$\boldsymbol{F}^e = \boldsymbol{K}^e \boldsymbol{d}^e$$

$$\boldsymbol{d}^e = [u_{1x}^e \;\; u_{1y}^e \;\; u_{2x}^e \;\; u_{2y}^e]^{\mathrm{T}}$$

$$\boldsymbol{F}^e = [F_{1x}^e \;\; F_{1y}^e \;\; F_{2x}^e \;\; F_{2y}^e]^{\mathrm{T}}$$



$$\cos\phi^{e} = \frac{x_2^e - x_1^e}{l^e} = \frac{x_{21}^e}{l^e}$$

$$\sin\phi^{e} = \frac{y_2^e - y_1^e}{l^e} = \frac{y_{21}^e}{l^e}$$

单元节点内力列阵 $\boldsymbol{F}^{e} = [F_{1x}^{e}\ \ F_{1y}^{e}\ \ F_{2x}^{e}\ \ F_{2y}^{e}]^{\mathrm{T}}$

单元节点位移列阵 $\boldsymbol{d}^{e} = [u_{1x}^{e}\ \ u_{1y}^{e}\ \ u_{2x}^{e}\ \ u_{2y}^{e}]^{\mathrm{T}}$

$$\boldsymbol{F}^{e} \overset{?}{\rightleftarrows} \boldsymbol{d}^{e}$$

在单元局部坐标系中：

$$\begin{bmatrix} k^e & -k^e \\ -k^e & k^e \end{bmatrix}\begin{bmatrix} u_{1x}^{\prime e} \\ u_{2x}^{\prime e} \end{bmatrix} = \begin{bmatrix} F_{1x}^{\prime e} \\ F_{2x}^{\prime e} \end{bmatrix}$$

$$\boldsymbol{F}^{\prime e} = \boldsymbol{K}^{\prime e}\boldsymbol{d}^{\prime e}$$

$$\boldsymbol{F}^{e} = \underset{?}{\boldsymbol{K}^{e}}\boldsymbol{d}^{e}$$

$$\boldsymbol{K}^{\prime e} = k^e\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$u_{Ix}^{\prime e} = u_{Ix}^{e}\cos\phi^{e} + u_{Iy}^{e}\sin\phi^{e},\ I = 1,2$$

$$\underbrace{\begin{bmatrix} u_{1x}^{\prime e} \\ u_{2x}^{\prime e} \end{bmatrix}}_{\boldsymbol{d}^{\prime e}} = \underbrace{\begin{bmatrix} \cos\phi^e & \sin\phi^e & 0 & 0 \\ 0 & 0 & \cos\phi^e & \sin\phi^e \end{bmatrix}}_{\boldsymbol{T}^{e}} \underbrace{\begin{bmatrix} u_{1x}^{e} \\ u_{1y}^{e} \\ u_{2x}^{e} \\ u_{2y}^{e} \end{bmatrix}}_{\boldsymbol{d}^{e}}$$

$$\boldsymbol{d}^{\prime e} = \boldsymbol{T}^{e}\boldsymbol{d}^{e}$$

$$u_{Ix}^{e} = u_{Ix}^{\prime e}\cos\phi^{e},\ u_{Iy}^{e} = u_{Ix}^{\prime e}\sin\phi^{e},\ I = 1,2$$

$$\boldsymbol{d}^{e} = \boldsymbol{T}^{e\mathrm{T}}\boldsymbol{d}^{\prime e}$$

$$\begin{aligned} \boldsymbol{F}^{e} &= \boldsymbol{T}^{e\mathrm{T}}\boldsymbol{F}^{\prime e} \\ &= \boldsymbol{T}^{e\mathrm{T}}\boldsymbol{K}^{\prime e}\boldsymbol{d}^{\prime e} \\ &= \underbrace{\boldsymbol{T}^{e\mathrm{T}}\boldsymbol{K}^{\prime e}\boldsymbol{T}^{e}}_{\boldsymbol{K}^{e}}\boldsymbol{d}^{e} \end{aligned}$$

$$\boldsymbol{K}^{e} = \boldsymbol{T}^{e\mathrm{T}}\boldsymbol{K}^{\prime e}\boldsymbol{T}^{e}$$

$$= \frac{k^e}{(l^e)^2}\begin{bmatrix} (x_{21}^e)^2 & x_{21}^e y_{21}^e & -(x_{21}^e)^2 & -x_{21}^e y_{21}^e \\ x_{21}^e y_{21}^e & (y_{21}^e)^2 & -x_{21}^e y_{21}^e & -(y_{21}^e)^2 \\ -(x_{21}^e)^2 & -x_{21}^e y_{21}^e & (x_{21}^e)^2 & x_{21}^e y_{21}^e \\ -x_{21}^e y_{21}^e & -(y_{21}^e)^2 & x_{21}^e y_{21}^e & (y_{21}^e)^2 \end{bmatrix}$$

单元应变：

$$\varepsilon^{e} = \frac{1}{l^e}[-1\ \ 1]\boldsymbol{d}^{\prime e} = \boldsymbol{B}^{e}\boldsymbol{d}^{e}$$

$$\begin{aligned} \boldsymbol{B}^{e} &= \frac{1}{l^e}[-1\ \ 1]\boldsymbol{T}^{e} \\ &= \frac{1}{(l^e)^2}[-x_{21}^e\ \ -y_{21}^e\ \ x_{21}^e\ \ y_{21}^e] \end{aligned}$$

单元应力：

$$\sigma^{e} = E^{e}\boldsymbol{B}^{e}\boldsymbol{d}^{e}$$

单元节点内力列阵

$$\boldsymbol{F}^e=[F_{1x}^e\ \ F_{1y}^e\ \ F_{1z}^e\ \ F_{2x}^e\ \ F_{2y}^e\ \ F_{2z}^e]^{\mathrm{T}}$$

单元节点位移列阵

$$\boldsymbol{d}^e=[u_{1x}^e\ \ u_{1y}^e\ \ u_{1z}^e\ \ u_{2x}^e\ \ u_{2y}^e\ \ u_{2z}^e]^{\mathrm{T}}$$

单元局部坐标系中的刚度方程：

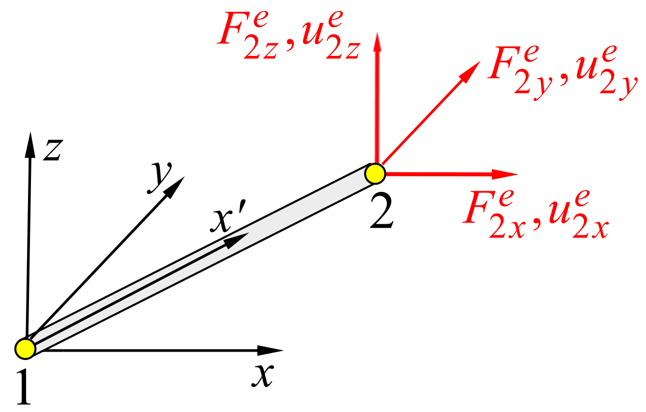
$$\underbrace{\begin{bmatrix}F_{1x}'^e\\ F_{2x}'^e\end{bmatrix}}_{\boldsymbol{F}'^e}=\underbrace{k^e\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}}_{\boldsymbol{K}'^e}\underbrace{\begin{bmatrix}u_{1x}'^e\\ u_{2x}'^e\end{bmatrix}}_{\boldsymbol{d}'^e}$$

全局坐标系中

$$\begin{aligned}\boldsymbol{F}^e&=\boldsymbol{T}^{e\mathrm{T}}\boldsymbol{F}'^e\\&=\boldsymbol{T}^{e\mathrm{T}}\boldsymbol{K}'^e\boldsymbol{d}'^e\\&=\boldsymbol{T}^{e\mathrm{T}}\boldsymbol{K}'^e\boldsymbol{T}^e\boldsymbol{d}^e\\&=\boldsymbol{K}^e\boldsymbol{d}^e\end{aligned}$$

$$\boldsymbol{K}^e=\boldsymbol{T}^{e\mathrm{T}}\boldsymbol{K}'^e\boldsymbol{T}^e$$

作业：推导三维杆单元的刚度方程和应力计算格式



$$u_{lx}'^e=\frac{1}{l^e}(x_{21}^e u_{lx}^e+y_{21}^e u_{ly}^e+z_{21}^e u_{lz}^e)$$

$$\begin{bmatrix}u_{1x}'^e\\ u_{2x}'^e\end{bmatrix}=\underbrace{\frac{1}{l^e}\begin{bmatrix}x_{21}^e & y_{21}^e & z_{21}^e & 0 & 0 & 0\\ 0 & 0 & 0 & x_{21}^e & y_{21}^e & z_{21}^e\end{bmatrix}}_{\boldsymbol{T}^e}\boldsymbol{d}^e$$

$$\boldsymbol{d}'^e=\boldsymbol{T}^e\boldsymbol{d}^e$$

$$\varepsilon^e=\frac{1}{l^e}[-1\ \ 1]\boldsymbol{d}'^e=\boldsymbol{B}^e\boldsymbol{d}^e$$

$$\begin{aligned}\boldsymbol{B}^e&=\frac{1}{l^e}[-1\ \ 1]\boldsymbol{T}^e\\&=\frac{1}{(l^e)^2}[-x_{21}^e\ \ -y_{21}^e\ \ -z_{21}^e\ \ x_{21}^e\ \ y_{21}^e\ \ z_{21}^e]\end{aligned}$$

$$\boldsymbol{\sigma}^e=E^e\boldsymbol{B}^e\boldsymbol{d}^e$$

2.2 单元刚度方程

单元刚度矩阵的物理意义

单元刚度方程给出了单元平衡时，单元结点内力和单元结点位移之间满足的关系式

$$\boldsymbol{F}^e = \boldsymbol{K}^e \boldsymbol{d}^e$$

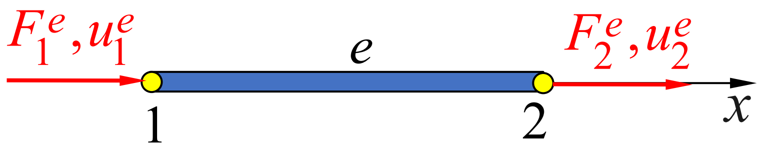
$$\begin{bmatrix} K_{11}^e & K_{12}^e & \cdots & K_{1m}^e \\ K_{21}^e & K_{22}^e & \cdots & K_{2m}^e \\ \vdots & \vdots & \ddots & \vdots \\ K_{m1}^e & K_{m2}^e & \cdots & K_{mm}^e \end{bmatrix} \begin{Bmatrix} d_1^e \\ d_2^e \\ \vdots \\ d_m^e \end{Bmatrix} = \begin{Bmatrix} F_1^e \\ F_2^e \\ \vdots \\ F_m^e \end{Bmatrix}$$

m — 单元自由度数

取 $d_j^e = 1,\quad d_i^e = 0(i \neq j)$

$$\begin{bmatrix} K_{1j}^e \\ K_{2j}^e \\ \vdots \\ K_{mj}^e \end{bmatrix} = \begin{Bmatrix} F_1^e \\ F_2^e \\ \vdots \\ F_m^e \end{Bmatrix}$$

当单元的第j个自由度给定单位位移而其它自由度固定时，为了使单元平衡而需要在单元各结点自由度上施加的结点力。



$$\boldsymbol{K}^e = k^e \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

K_{ij}^e的物理意义：当单元的第j个自由度给定单位位移而其它自由度固定时，为了使单元平衡而需在单元第i个自由度上施加的结点力。

单元刚度矩阵的特性

$\boldsymbol{F}^e = \boldsymbol{K}^e \boldsymbol{d}^e$

- 对称性

  $\boldsymbol{K}^e = \boldsymbol{K}^{e\mathrm{T}}$

- 对角元非负性 K_{ii}^e

  取 $d_i^e \neq 0, d_j^e = 0 (j \neq i): F_i^e = K_{ii}^e d_i^e$

  对于稳定的物理问题，F_i 和 d_i 的方向应该相同 $\Longrightarrow K_{ii}^e > 0$　　$K_{ii}^e = 0$?

- 奇异性 $|\boldsymbol{K}^e| = 0$ ？
  - 单元自由：即使单元平衡，方程 $\boldsymbol{K}^e \boldsymbol{d}^e = \boldsymbol{F}^e$ 也不存在唯一解（存在刚体位移）
  - 每一列都要满足平衡条件

    一维杆单元：$K_{1j} + K_{2j} = K_{j1} + K_{j2} = 0; \quad j = 1,2$

    当单元的第j个节点位移为1而其它节点位移为0时，施加在单元上的节点力在x方向的平衡条件

- 二维和三维问题中刚度阵各行和各列有何关系？

贝蒂互易定理

$$\boldsymbol{d}_1^{e\mathrm{T}} \boldsymbol{F}_2^e = \boldsymbol{d}_2^{e\mathrm{T}} \boldsymbol{F}_1^e$$

$$\boldsymbol{d}_1^{e\mathrm{T}} \boldsymbol{K}^e \boldsymbol{d}_2^e = \boldsymbol{d}_2^{e\mathrm{T}} \boldsymbol{K}^e \boldsymbol{d}_1^e$$

$$= \boldsymbol{d}_1^{e\mathrm{T}} \boldsymbol{K}^{\mathrm{T}} \boldsymbol{d}_2^e$$

$$\boldsymbol{d}_1^{e\mathrm{T}} (\boldsymbol{K}^e - \boldsymbol{K}^{e\mathrm{T}}) \boldsymbol{d}_2^e = 0$$